



COVID-Vis: Visualizing knowledge exchange on scientific software development in the COVID-19 era

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ABSTRACT

As the world is still recovering from the detrimental effects of the COVID-19 pandemic, one key aspect of the pandemic era were the global efforts for containment, case tracking and several other factors. While the scientific and governmental initiatives were largely successful and effective, a notable surge was observed in contributions from individuals and programming communities that developed their own software for COVID-19 by using data retrieval and analysis along with visualization methodologies. To achieve their goals, they turned their attention to knowledge exchange portals and asked questions regarding technological queries. In this paper, we present a collective platform that retrieves such questions from a well-known Q&A portal and visualizes the contained information. This platform serves as a useful tool for assessing programming and technological interest in COVID-19 related software development efforts while also promoting the open science principles.

CCS CONCEPTS

• **Information systems** → World Wide Web; Web interfaces; Information retrieval; • **Human-centered computing** → Visualization; Visualization systems and tools; • **Software and its engineering**;

KEYWORDS

COVID-19, Knowledge exchange, Data Visualization, Data Retrieval, Platform

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1 INTRODUCTION

The COVID-19 pandemic was undoubtedly a defining crisis of the last decade, impacting every functional aspect of society and disrupting well-established operations [1, 2]. As the severity of the virus was rendering its containment, cure and prevention via vaccinations urgent, the existing scientific, academic and governmental communities in a global scale strived to provide practical solutions in a global scale [3-5]. In parallel, in an effort to democratize the process of developing pandemic related solutions, many organizations and initiatives supported the open science principles during COVID-19, encouraging the public sharing of produced data and promoting open-source software [6, 7]. Indeed, apart from official sources, a wide variety of open and public repositories of knowledge regarding COVID-19 is observed, with subjects relevant to data mapping [8], accumulation of data sources [9], case tracking [10], to name a few. It is thus evident, that open science contributes to sensitizing the public in regard to COVID-19 and its effects, while also multiplying the initiatives undertaken by various individuals for processing available data.

An indicative case of this statement is the growing interest from individuals in leveraging openly available data and specific technologies and tools in order to conduct their own projects related to COVID-19. These individuals, ranging from experienced developers to amateur data enthusiasts, turned their attention to knowledge sharing communities, such as Stack Overflow (SO), in order to seek answers to their technological queries [11, 12]. These questions, contain various different technologies utilized in COVID-19 solutions and comprise a valuable repository of knowledge, that showcases the overall interest of the public in the pandemic and in Scientific Software Development (SSD). Thus, the motivation behind this study was to create a dynamic dashboard that would accumulate the data of COVID-19 questions from SO and visualize their geographical, temporal and technological evolution.

To that end, we present **COVID-Vis**, a collective dashboard of SO data that cleverly presents and visualizes the technological trends extracted from SO data, during COVID-19. More specifically, COVID-Vis comprises an efficient platform that aims to:

- A) **Accumulate** data of multiple forms (geographical coordinates, unstructured text, numerical);
- B) **Store and analyze** the collected data;
- C) **Visualize** the data using appropriate tools and packages;
- D) **Present** the data in an efficient manner.

The **COVID-Vis** platform is based on our previous works [11, 13] on the same subject and contributes to the current developed solutions for COVID-19 by offering an alternative perspective on the race for studying and analyzing the pandemic. More specifically, it highlights prominent technologies that are utilized for the analysis of COVID related data, indicating technological directions for future projects and providing a roadmap of technological demand during the pandemic era.

2 RELATED WORK

There has been a plethora of academic research on knowledge exchange in Q&A portals, as they constitute a valuable source of user interactions and technological advancement [14, 15]. In the context of this study, SO has been thoroughly analyzed from various domains in regard to the content that it provides. Here we provide some indicative works but there is an extensive pool of research papers on the matter.

As SO is a valuable source of information on multiple different technological domains, many studies have performed empirical analyses by collecting user discussions on specific fields and extracting insights and topics. As the fields of IT are developing, there have been empirical studies on concurrency and security [16], machine learning applications [17] and libraries [18] as well as mobile development [19]. In addition, other works focus on security and privacy questions [20, 21] while IoT [22], API recommendation [23] and continuous software development [24] also captivate the interest of researchers. Some studies also explore the effect on crowdsourcing in SO and how do users interact as an interconnected community [25–27]. In addition, as tags are an important property of SO questions that indicate technological objectives, Chen et al. utilize them to detect associations between technologies [28] while the concepts of tag hierarchies and groups [29] as well as tag prediction [30] are also explored.

Furthermore, several studies explore the questions that are being posted both in terms of writing guidelines [31] but also on their characteristics, such as code snippets and bounties [32, 33]. Barua et al [34] perform a large-scale study on questions in order to detect trends and topics, while Beyer et al. [35] define question types that are being posted, ranging from questions directed towards learning to questions for concepts and methods. Zhu et al. focus on question discussions and response times [36] while Correa et al [37] study the criteria of closed questions and their quality. Finally, many studies concentrate on answer characteristics and the factors that affect the possibility of a question receiving answers [38–41].

In a more practical and pandemic related level, multiple works from governmental, scientific and open science bodies have led to the development of notable platforms related to the COVID-19 pandemic. While each country has developed their own case tracking and vaccination platforms, there have been collective platforms launched by WHO [45], the John Hopkins University [46] and CDC [47] that aim at providing a clear picture of the global status of the pandemic and roadmaps for future measures and steps. In addition, there have been efforts in promoting open science to combat the pandemic both in research works [42, 43] and in platforms with informative material [48]. Finally, independent communities

and organizations are actively engaging in transparent data sharing and portal creation with interactive pandemic maps [49] and openly available datasets [50]. Generally, open science is hailed as a torchbearer in addressing crises and is actively encouraged by high profile organizations [51], committees [52] and European bodies [53].

3 SOFTWARE DESIGN & DEVELOPMENT

3.1 Usefulness of Software – Use cases

As mentioned in the Introductory section, the development of COVID-Vis aimed at addressing the lack of organized platforms that collect data related to technological trends and knowledge exchange during the pandemic. That, in turn, would reveal the effect that COVID-19 had to the world of scientific software development and the community of SO. By providing users with the features and capabilities of COVID-Vis, we facilitate interested parties and groups in acquiring insights on the matter and advance their scientific purposes or simply be up to date. To better illustrate the multiple different parties that could use the platform to their benefit, we provide some indicative examples of use cases below. Hence, possible uses of the platform are:

A) A researcher that wishes to write a scientific paper regarding subjects of technological analysis or trends during the COVID-19 era. This researcher would require insights on the effect of the pandemic on technological communities and programmers in Q&A portals are a perfect example of such a community. Thus, the researcher could enter the platform and acquire knowledge on the most popular COVID-19 scientific software development questions, find prominent technologies and users as well as track the countries that are most interested in contributing on the technological battle against the pandemic.

B) From an educational perspective, an academic educator could leverage the platform to showcase a case study on the changes that the pandemic brought on multiple fronts. The professor could begin a lecture with a general observation on the effect that the pandemic had in every aspect of society and gradually lead the students in the technological and open science sector. Then, by using the platform, a brief introduction in changing technologies during the pandemic can be demonstrated along with fruitful discussion on knowledge sharing communities and portals. In addition, by observing the most utilized technologies, he/she could gain valuable insights or ideas and integrate them in their teaching materials.

C) Stakeholders in corporations or companies that are curious about the contribution of open science and user interactions during the pandemic from a discrete target group.. Given that, apart from government efforts, there are many tools developed from common enthusiasts, stakeholders can understand what technologies these groups prefer and possibly suggest some business strategies for more efficient product development.

D) From the perspective of a simple user, this platform is a very useful tool that provides easily digestible knowledge on the subject of technological trends in the COVID-19 era without wasting time in confusing visualizations. All the necessary data are presented in one collective platform via the use of easily interpretable graphs, facilitating the informing of simple users on this highly important topic.

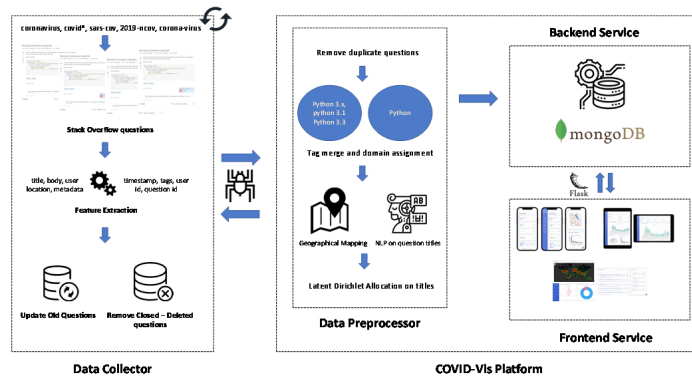


Figure 1: Platform Architecture

3.2 Software Architecture

The developed platform is divided into several components, each with a specific operation that complements the design. Each component operates separately but also in collaboration with the others in order to provide the collective platform service. In Figure 1 we provide an overview of the architecture.

More specifically, the separate components of the platform are:

The Data Collector: This component serves as an intelligent agent that continuously scrapes SO for questions regarding software, tools and general technological tasks developed for COVID-19. To achieve this, the constructed data scraping service accesses the SO website and queries the available questions based on a constructed search string. The search string consists of keywords related to COVID-19 (coronavirus, covid*, sars-cov, 2019-ncov, coronavirus). The developed service runs once a week and collects all existing questions that correspond to this search string. In addition, apart from collecting new questions, the Data Collector service updates old questions with new values on several metadata that may change (e.g. votes, views) and deletes questions that have either been marked as closed by SO, meaning that they do not have scientific value and been covered by another question, or questions that have been deleted. The features that are parsed and collected are the timestamp of the question, the Id of the user that posted the question, the accompanying tags that refer to the technologies associated with the question, the question body and title, the user location as well as several metadata (question votes, question views, question ID). At the time of this paper’s completion, the platform contains 4189 unique questions, posted by 3471 different users.

The Data Preprocessor: The collected questions are then passed into a different component that is charged with performing necessary preprocessing procedures and convert the extracted features to valuable information. First, the preprocessor removes duplicate questions based on the question ID. Next, the geographical information is converted from text to coordinates via specialized Python Packages while the tags of each question are distributed to a predefined lexicon of technology domains utilized in our previous works [11, 13]. Thus, apart from the accompanying tags, each question is mapped to one or more technological domains that represent its broader technological scopes. It should be noted that similar

tags (e.g. python 3.1, python 3.x) are merged into one identical tag (e.g. python). The textual information is also subjected to Natural Language processing techniques (tokenization, stopwords, stemming) to represent it as a vector. In addition, the Latent Dirichlet Allocation topic modelling methodology is implemented, in a similar manner to our previous works in order to draw the thematic axes of the posted questions and visualize them on the platform.

The Backend Service: Once the Data Preprocessor has completed all the tasks, the preprocessed features along with the collected metadata are stored in the Backend Service. The constructed Backend Service consists of a Mongodb database that stores the collected questions as a list of features. In addition, the Flask framework is employed to ensure the communication between the Backend and Frontend Services in order to transfer and visualize information on the platform.

The Frontend Service: This component is the final component of the platform and the one that users are able to see when they use the platform. It is a fully responsive interface designed with standard languages (HTML, CSS, Javascript) and auxiliary frameworks (e.g. Bootstrap). We have invested in an easily interpretable UX design that favors simplicity and allows the user to understand the purpose of the platform, without being overwhelmed with redundant information. The developed platform produces several graphs and visualizations that portray user engagement, technologies and thematic circles while also ensuring dynamic capabilities by allowing users to specify different time periods and view analyzed questions on different timeframes for a clearer view of trends.

4 SOFTWARE DEMONSTRATION

In this section we present the frontend service of the platform, with all the available question data. We highlight each section of the platform, while at the same time we discuss the findings and what they represent regarding the increase of open science efforts during COVID-19.

In Figure 2 we present an overview of the platform, showcasing all the available functions, as well as showing its responsiveness in mobile phones and tablets. A more detailed video presentation can be found here [54]. At the top, we have the Platform State Data Cards (No.1), which provide a useful summary of the basic features



Figure 2: Platform Overview and Responsiveness

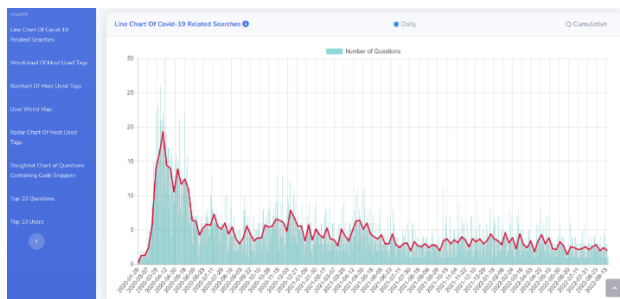


Figure 3: Linechart of Temporal Information

of the platform, including total number of questions, users and the timestamp of the last update. The cards also provide average values of the collected metadata. By the time of writing, each posted question has received an average of close to 2 comments, 1 answer and 1 vote. Moreover, the platform offers the possibility of altering the dates according to the question timestamps and view specific data from different time windows, offering a more comprehensive look on software development trends for COVID-19. Directly below, there is a comprehensive linechart of the daily posted questions along with an available cumulative option. As seen in Figure 3, the interest in developing technological solutions and preventive measures related to the pandemic peaks during the spring and summer of 2020 were beginning to take form. Subsequent periods show some increasing trends, though the general decrease in cases and severity has led to a drop in recent months.

In order to profile the technological interests of users that post questions in SO, we provide a wordcloud and a bar chart of the most widely used tags (Figure 4a), which express technologies. Based on the demonstration of the platform, at the time of writing the most utilized technologies appear to be data manipulation and analysis languages (e.g. python, r, javascript), along with libraries and frameworks for data analysis and visualization (pandas, reactjs, django, matplotlib, ggplot2). In addition, the importance of data collection

is highlighted, with the use of specialized tools (beautifulsoup, selenium, json) in combination with machine learning approaches for the development of case prediction models (tensorflow, keras). To further profile the technological aspect of the questions, a radar chart was created to express the most popular technological domains, in line with our previous work [11, 13]. Via the radar chart we can see that most questions revolve around the technological domain of Languages, with Web Frameworks and Platforms occupying the following higher spots. This proves that the open science community, in its efforts to support the governmental and societal struggle with COVID-19, relied on IT technologies that facilitated the development of scientific software. In addition, individuals that post their SSD related questions on knowledge exchange portals seem eager to seek advice on code related problems, as 90% of the posted questions contained code snippets (Figure 4b).

The geographical mapping of the countries that represent the users who support open science for COVID-19 and contribute to its efforts (Figure 4c) reveals that the most active groups of users come from European countries, India and the United States of America. Other interesting observations are linked to the presence of active users in Indonesian countries, Argentina and other South American countries (e.g. Peru, Venezuela) and Australia. The fact that different users from different countries come together to fight a common enemy is an indicator of the power of technological knowledge exchange and proves that global collaboration can lead to fascinating outcomes.

Finally, the application of the LDA [44] algorithm on the question texts is depicted in Figure 5. The size and divergence between the extracted topics have been projected on a two-dimensional space, via the use of multidimensional scaling while in the right part of the figure the main words that characterize each topic are presented. For example, we can see that Topic 1 can be interpreted as “Data Manipulation, Analysis and Visualization” due to the existence of several words that point to this direction (e.g. python, data, plot, map, graph etc.). The radius of each circle indicates the most prevalent topics in questions, which are the central points of discussions, while the distance between the circles showcases the similarity or dissimilarity between topics. Evidently, as the platform

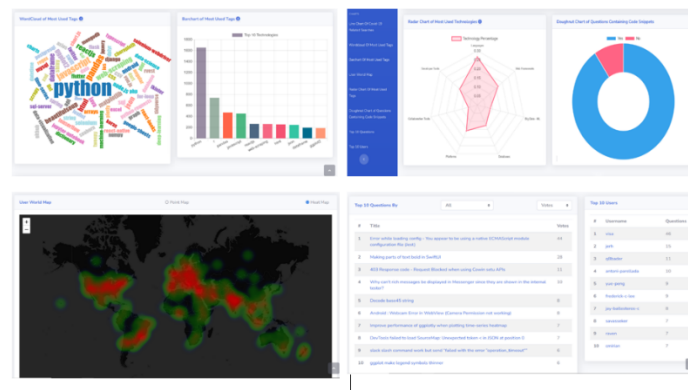


Figure 4: Platform Visualizations for (a) Technologies, (b) Technological Domains and Code Snippets, (c) Geographical mapping and (d) Sorting of questions based on metadata

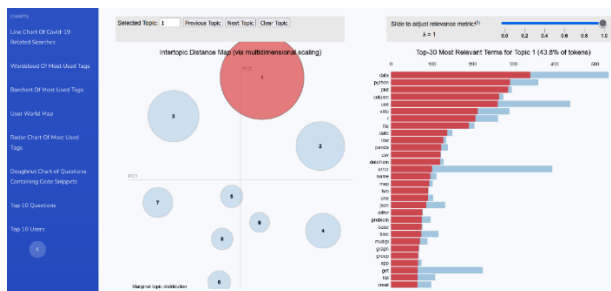


Figure 5: LDA visualization of COVID-19 question topics

is dynamically updated and more questions are added or deleted, the topics may change, according to discussion trends.

5 CONCLUSIONS

The goal of this paper was to thoroughly present a comprehensive and interpretable platform that visualized the knowledge exchange between individuals who wished to join the technological battle against a formidable disease and provide software solutions to assist the overall efforts. The collected data from a well-known Q&A portal indicate that an impressive number of users amassed and posted queries for improving and developing their solutions, while the technical and textual aspects of their questions reveal interesting insights about the prominent technologies and topics of scientific software development and open science during this turbulent period.

The prime technologies in the investigated period seem to refer to programming languages aimed at data analysis and development (e.g. Python, R) along with frameworks (Reactjs) as well as data related tasks such as geographical mapping, visualizations and machine learning models. Concurrently, the thematic axes complement the technological aspects and are relevant to data manipulation or other practices. Finally, the overall number of questions, users and general engagement in discussion topics constitute a clear example of a united community that came together to provide knowledge and respond to a global crisis. This is an encouraging sign for future

phenomena, indicating the readiness of the programming landscape and the willingness to provide support.

From an ICT perspective, the primary research contributions are the clear tracking of emerging technologies and tools that facilitate data analysis, web development and software engineering. In addition, we provide a roadmap on developer preferences in ICT software, based on popular demand and knowledge exchange. The implications to wider groups and researchers are the reflection of societal support in a time of crisis, by using cutting-edge technologies, and the encouragement to adopt these practices on other scientific fields. Furthermore, the development of the platform promotes open science practices and open-source software principles. Interesting future work directions for the current study can be the inclusion of more Q&A portals such as Reddit or Quora, software quality metrics and GitHub repositories related to COVID-19 SSD as well as the integration of personality profiling and characteristics along with sentiment analysis of the available users.

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